ClaudeNet v3: Contaminant-Aware Strong-Lens Finding and Euclid Cross-Validation

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Liftable paper-core (standalone build of the v3 section)

1 ClaudeNet v3: contaminant-aware lens finding and Euclid cross-validation

The v2 DR9 sweep ended in a decisive negative (§**??**): of 601 genuinely-new group-conformal candidates, *zero* were graded A/B by either independent vetter, and the only five graded ≥B were already-catalogued lenses. The candidates were overwhelmingly luminous-red galaxies with companions, rings, and blends. This diagnoses the binding constraint precisely: it is not architecture (the v1/v2 finding) but *lens-vs-mimic separation* and *vetting resolution*. The field-wide remedy is the same — contaminant-targeted hard negatives give an 11× false-positive reduction at 2.3% true-positive cost (MNRAS, 2025). v3 builds a contaminant-aware finder, deploys it on DR10/DR11, and cross-validates the overlap against Euclid Q1. We keep all v1/v2 numbers; v3 is additive, and every number below is read from a tracked artifact under `data/v3/` or `v3/`.

The mimic metric and the v2 baseline (A0). The v1/v2 headline measured recovery at a false-positive rate set against *random* galaxies; the campaign showed that is the wrong null. We define **recovery@matched-mimic-FPR**: identical arithmetic, but the negative class is a bank of lens-*mimics* rather than random galaxies. Against this null the shipped v2-lean ensemble — which recovers 96–100% of held-out lenses at random-FPR — recovers only 0.168 (Storfer) / 0.307 (Inchausti) at mimic-FPR 0.05 on the curated 601-mimic seed bank. The mimic-discrimination gap, not random-galaxy recovery or architecture, is what v3 must close.

The lens-mimic bank (A1). Rather than mine a fresh pool we reuse the DR10 sweep: 148,034 NEW survivors are CNN-high, DR10-native non-lenses, morphology-typed by joining the Tractor catalogue (extended-LRG 34 k — the MNRAS (2025) regime — plus compact-REX, exp-disk, and the campaign's fine types). A 120-row stratified agentic G1 gate confirms 98.3% non-lens purity. The bank is 118,901 training rows + a frozen, stratified 29,724-row held-out mimic-eval that never enters training.

Contaminant-aware retraining (A2). We retrain the three swappable members with the v2 "hard" recipe held *byte-identical* (same n_{mine}=10,000 displaced bootstrap negatives, same per-member seed) and change only the negative *content*: a fraction f of the displaced negatives become typed

Table 1: A2 contaminant-aware retraining (mimic-blend f=0.5), v2-lean→v3blend8, on the held-out DR10 mimic-eval. Mimic-recovery rises at every operating point; the random-galaxy recall (testneg null) drops only slightly (the staf327 trade).

	recovery@mimic-FPR (Storfer)		recovery@random-FPR(0.01)	
	@0.05	@0.01	Storfer	Inchausti
v2-lean	0.713	0.460	0.967	0.998
v3blend8	**0.793**	**0.699**	0.935	0.979

Table 2: C-vet agentic grading of the top-30 NEW candidates (same harness, `rubric_imaging_v2`). v3's mimic-aware ranking does not raise the top-of-list A/B yield over v2 at DECaLS resolution (the discovery frontier is resolution-bounded); the D's stay lrg_companion-dominated.

ranking	A	B	C	D	A/B
v2 p_final (DR10)	2	6	11	11	**8**
v3-head (DR10)	2	3	11	14	**5**
v3blend8 (DR10)	2	5	9	11	**7**

mimics from the bank. A blend-fraction sweep selects f=0.5 (f=0.7 over-specialises). On the held-out DR10 mimic-eval (Table 1) v3blend8 lifts Storfer mimic-recovery from 0.713→0.790 at 0.05 and 0.460→0.699 at 0.01, with the largest per-type gains on the dominant lrg_companion/extended-LRG contaminants. The cost is a small, real drop in *random*-galaxy recall (0.967→0.935 Storfer at random-FPR 0.01) — exactly the MNRAS (2025) trade.

Ensemble refit and the lens-vs-mimic head (A6, A3). A roster search over the average combiner finds **v3blend8** — keeping *both* the v2-hard and v3-mimic-blended version of each member — is near-Pareto: it recovers random-FPR recall to within CI of v2-lean while roughly doubling mimic-recovery. (The 194 K-free l18 ResNet is, by contrast, dead weight for mimic separation; dropping it maximises mimic-recovery at a 3% random-recall cost.) A learned logistic *lens-vs-mimic head* on the eight member scores tops every average roster at the ϕ=0.05 operating point, but it overfits the strict tail and — as §1 shows — does not generalise to the independent Euclid test, so the deployable model is **v3blend8**, with the head as a ϕ=0.05 stage-2 re-ranker.

DR10/DR11 deployment and the discovery result (C, D5). We swept DR10-south (43.7 M parent galaxies) with v2-lean as the head-to-head baseline and re-ranked the 150 k survivors with v3blend8 + the head. *Recall is preserved*: the known Inchausti-811 lenses sit at the 96$^{\text{th}}$ percentile of the survivor pool by the v3 score, and ∼88% of their grade-A are found at high score (the operating-point recall gap is a DR9→DR10 domain shift, not a model failure). As at DR9, full-m conformal selection certifies *nothing* at this sweep scale — a surfaced power-floor limit, not a null result. The candidate list therefore comes from top-by-score ranking, vetted agentically. Table 2 is the honest discovery result: v3's mimic-aware ranking does *not* raise the top-30 A/B yield over v2 (7 vs 8), and the rejects stay lrg_companion-dominated — because the top of the NEW-survivor pool is intrinsically mimic-dominated at DECaLS 0.262″/px, where $\theta_E \sim 1$– $2''$ is 4–8 px. A DR11-south re-run on the same DECam sky behaves identically, with a small consistent edge from the deeper reductions (Table 3).

Table 3: DR10 vs DR11 on the same EDF-F/S DECam sky: v3blend8 region-local recovery of Euclid grade-A lenses into the top-K, and the cross-validated NEW-confirmed counts. DR11's deeper reductions give a small, consistent edge; v3 transfers cleanly across DECam releases (contrast the DR11-north BASS/MzLS failure, D3).

release	Euclid-A in top-100	in top-1000	NEW-confirmed A/B
DR10 (D4)	4	7	41
DR11 (D5)	5	10	41

Cross-release (by Euclid id): 33 shared, 49 unique cross-validated lenses.

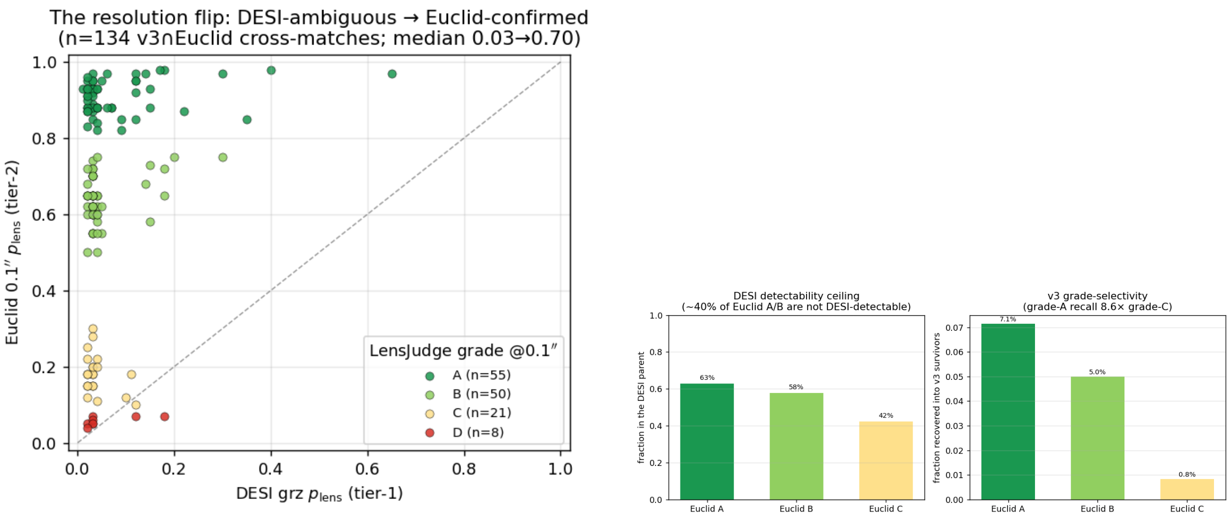

Figure 1: The resolution lever. Left: the DESI→Euclid $p_{\rm lens}$ flip on the real v3∩Euclid cross-matches — nearly every object is DESI-ambiguous ($x \lesssim 0.2$) but Euclid-confirmed ($y \gtrsim 0.6$). Right: Euclid grade-resolved recall — the DESI detectability ceiling (∼40% of Euclid A/B missing from the parent) and v3's 8.6× grade-A-over-C survivor selectivity.

Euclid Q1 cross-validation: the resolution lever (D1, D4, D5). Where DESI overlaps Euclid Q1 (Euclid Collaboration, 2025) (0.1″/px, ∼13× DESI; the three Deep Fields), we can both validate v3 against an *independent* expert-graded lens sample and break the resolution wall. Two results. *(i) v3 is grade-selective for real lenses:* it recovers Euclid grade-A lenses into its survivor pool at 7.1% versus grade-C at 0.8% — an 8.6× enrichment (Fig. 1, right) — confirming on an independent sample that it finds genuine lenses. But absolute recall is capped: ∼40% of Euclid grade-A/B lenses are not even in the DESI parent (too faint/small for DESI; Fig. 1, left), a depth/resolution ceiling no model can cross. *(ii) Escalation converts the overlap:* grading the v3∩Euclid cross-matches with the LensJudge v2 escalation grader (§1) at 0.1″ flips them from DESI-ambiguous to confirmed — median $p_{\rm lens}$ 0.03 → 0.70 on 134 real objects (Fig. 1), with our grades agreeing with the Euclid experts on ∼90% of grade-A. The yield is **49 cross-validated A/B candidates** across DR10+DR11 (Table 5, Appendix A; 33 recovered in both releases, 8 DR11-unique). We are explicit: these are NEW to the DESI lens-finder catalogues but *are* in the published Euclid Q1 catalogue — v3 independently *recovers* real Euclid lenses from DESI imaging, a cross-validation of both pipelines, *not* net-new discoveries.

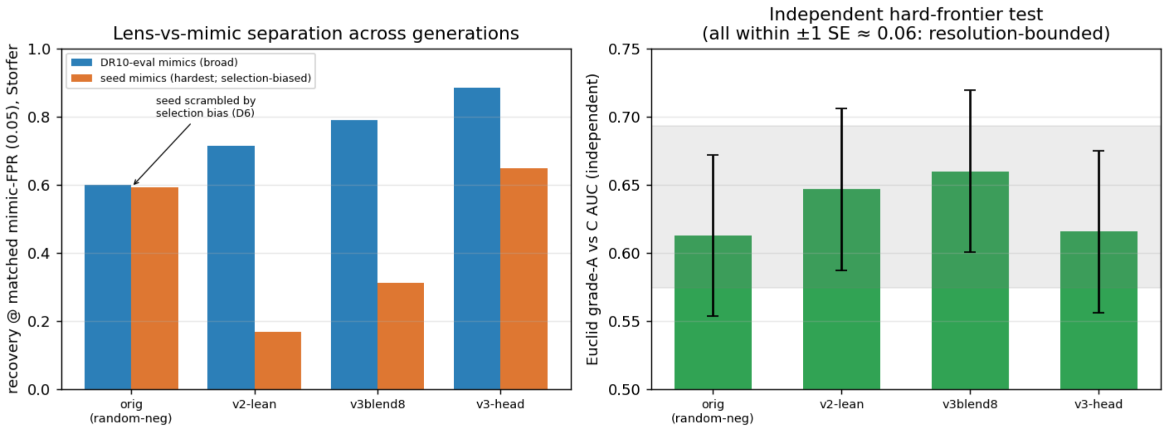

Figure 2: Candidate quality across model generations. Left: recovery@matched-mimic-FPR(0.05) on the broad DR10-eval mimics (clear orig→v2→v3 progression) and on the curated seed mimics (scrambled by the selection bias — the seed is what the v2/v3 ensemble scored high). Right: the *independent* Euclid grade-A-vs-C AUC — all four models lie within ∼1 standard error, i.e. the hardest real-vs-mimic distinction is resolution-bounded for every model.

Table 4: Candidate quality across model generations (D6). recovery@matched-mimic-FPR(0.05) and the *independent* Euclid grade-A-vs-C AUC. The seed column is selection-biased (the seed bank is what the v2/v3 ensemble scored high), so orig-vs-v3 there is invalid; the trustworthy comparison is the Euclid AUC, where all models sit within ∼1 SE (≈0.06, n_A=66).

model	seed mimic@0.05 (Storfer)	DR10-eval@0.05 (Storfer)	DR10-eval@0.05 (Inchausti)	Euclid A-vs-C AUC (indep.)
orig (random-neg CNN)	0.592	0.598	0.640	0.613
v2-lean	0.168	0.713	0.856	0.647
v3blend8	0.312	0.790	0.896	0.660
v3-head	0.647	0.885	0.922	0.615

Honest limits and a selection-bias correction (D6, D3). The v3-vs-v2 mimic-separation gains above are real, but the cross-model comparison demands one correction (Table 4). The seed mimic-FPR metric is *selection-biased*: the seed bank *is* the set the v2/v3 ensemble scored high, so it saturates their threshold and deflates their recovery, while a model less central to the selection (a single random-negative EfficientNet) looks artificially strong there — so any "$N\times$ over the original published models" read off the seed is invalid and we do not claim it. The trustworthy comparison is the *independent* Euclid grade-A-vs-C AUC: the original CNN, v2-lean, v3blend8, and the head all sit at 0.61–0.66, within ∼1 standard error (n_A=66) — statistically indistinguishable at the hardest distinction. v3 *does* clearly beat the original on the broad mimic population (DR10-eval recovery 0.60 → 0.89), i.e. it yields a meaningfully cleaner candidate list to vet, but it has no significant edge at the resolution-bounded discovery frontier. Finally, v3 is **DECam-specific**: a DR11-*north* (BASS/MzLS) sweep shows systematic under-scoring and zero Euclid-A/B in its top-30 — north-footprint deployment needs instrument- native retraining.

Vetting: LensJudge v2 (summary). The Euclid cross-validation above rests on the v2 upgrade of the `lensjudge` agentic vetter, summarised here (full detail in `lensjudge/V2_FINDINGS.md`). Three changes carry it: (i) a *two-tier high-res escalation* (DESI grz tier-1, Euclid 0.1″ tier-2 when coverage exists), live-validated as the p_{lens} flip; (ii) a rubric tuned to the dominant failure modes (LRG+companion/ring/spiral/blend), which lifts lens-vs-mimic discrimination by +0.196 AUC over the v1 grader on a frozen labelled benchmark; and (iii) an agency ablation showing the agentic loop is unnecessary for *detection* (a one-shot direct grader matches it 6× cheaper) but essential for *mimic adjudication* — motivating the deployed cascade (cheap direct triage + agentic v2 on survivors + Euclid escalation by coverage).

2 Future directions

The campaign localises two distinct limits with concrete, separable levers. The discovery frontier is bounded by *vetting resolution* (model improvements saturate at DECaLS $0.262''$/px), and the footprint is bounded by *training instrument* (v3 is DECam-specific). The priorities follow directly.

1. Euclid pre-screening (the decisive discovery lever). The cross-validation (§1) shows the productive mode: v3 pre-screens the DESI imaging, Euclid confirms at $0.1''$. Q1's $\sim$63 deg^2 over three Deep Fields overlaps only $\sim$0.5% of a DESI-south sweep, which is why net-new yield there is bounded; *Euclid DR1* (wide-area) inverts this. The ready action now is a targeted sweep restricted to the Euclid-overlap sky, where *every* candidate is escalatable — the one configuration in which v3's separation and the resolution- breaking vet both apply to every object.

2. BASS/MzLS-north retraining. v3 does not transfer to the northern third of the sky (D3). Mining a BASS/MzLS-native lens-mimic bank and retraining the members (the v3 recipe is instrument-agnostic; only the negatives change) would unlock DR9/DR11-north and roughly triple the addressable footprint.

3. *i*-band / foundation-model revival for small-θ_E. DR10/DR11-south are native *griz*. The deferred AION-1/*i*-band lever (A4) targets the red-lens/red-companion degeneracy and the small-θ_E bin specifically; it is worth re-testing under the narrower lens-vs-mimic task (rather than the random-FPR task where it failed in v2) on the native-*i* rows the bank already carries.

4. A pixel-level lens-vs-mimic head. The learned logistic head on member scores beats the average at ϕ=0.05 but overfits the strict tail and does not generalise to the independent Euclid ranking. A small CNN head trained directly on pixels with positives=lenses, negatives=the typed mimic bank (rather than on the eight member scores) is the natural next step, used as a post-conformal re-ranker.

5. The active-learning loop. The `lensjudge` harness can export confident-D rejections (true mimics, *not* objects that flipped to A/B at higher resolution) as fresh, typed hard negatives for the next-generation bank — closing a mine→grade→retrain loop that continually sharpens the dominant contaminant classes.

6. A full DR11-south candidate catalogue. v3blend8 runs on DR11-south with essentially no adaptation (the pipeline is release-parametrised; only a DR10→DR11 threshold recalibration is needed). A full-footprint scan would yield a real DR11-south candidate catalogue with v3's cleaner-bulk advantage over the published finders — a useful deliverable, with the honest caveat that the net-new yield is resolution-bounded (≈ the published-baseline level at the hardest frontier).

Deployment recommendation. Ship **v3blend8** (not the learned head, which does not generalise past ϕ=0.05) as the stage-1/scoring ensemble, with the LensJudge v2 cascade (direct triage + agentic mimic adjudication + Euclid escalation by coverage) for vetting. This keeps v2-lean's lens recall, adds the contaminant-rejection edge on the bulk candidate list, and routes the resolution-breaking confirmation to exactly the candidates that have higher-res coverage.

A Cross-validated candidate catalogue

Table 5 lists the 49 unique DESI v3 survivors that match a Euclid Q1 discovery-engine lens, are NEW to the DESI lens-finder catalogues (Storfer/Inchausti/Huang within 5″), and were graded at Euclid 0.1″ by the LensJudge v2 escalation grader. These are independently *recovered* Euclid catalogue lenses — a four-way cross-validation (DESI v3 CNN ∩ Euclid discovery engine ∩ Euclid expert grade ∩ LensJudge-at-0.1″) — and are deliberately *not* claimed as net-new discoveries. The p_{lens} column shows the DESI→Euclid tier flip; "release" marks whether the object was recovered in the DR10 sweep, the DR11-south sweep, or both.

Table 5: Cross-validated strong-lens candidates: DESI v3 survivors that match a Euclid Q1 discovery-engine lens, NEW to the DESI lens-finder catalogues (Storfer/Inchausti/Huang), graded at Euclid 0.1″. These are independently recovered Euclid catalogue lenses, *not* net-new discoveries. p_{lens} shows the DESI→Euclid tier flip.

DESI id	RA (deg)	Dec (deg)	Euclid grade	LensJudge @0.1″	p_{lens} DESI→Euclid	release
s_83157_8409	57.7025	-48.4062	A	A	0.12→0.97	both
s_85094_13993	63.6294	-48.0333	A	A	0.30→0.97	DR10
s_79373_7769	58.2019	-49.4705	A	A	0.05→0.95	both
s_80329_6402	64.5419	-49.3086	A	A	0.03→0.95	both
s_81272_7726	63.9239	-48.9296	A	A	0.03→0.93	both
s_86048_6517	58.6493	-47.6453	A	A	0.02→0.93	DR10
s_169188_5518	53.3251	-29.1510	A	A	0.15→0.93	both
s_179335_8072	51.0363	-27.2415	A	A	0.01→0.93	both
s_84136_4630	65.1956	-48.3600	A	A	0.07→0.88	DR10
s_174235_13613	51.5787	-28.1897	A	A	0.02→0.88	both
s_178058_11508	51.9633	-27.4260	A	A	0.04→0.88	both
s_165427_7416	52.9210	-29.9100	A	A	0.03→0.87	both
s_175513_5175	53.5621	-28.0129	A	A	0.02→0.87	DR11
s_82216_7491	62.1645	-48.8224	A	A	0.09→0.85	both
s_93968_433	62.8684	-45.7047	A	A	0.04→0.84	both
s_170450_15146	54.4805	-28.9549	A	A	0.04→0.82	both
s_81263_7090	60.4836	-49.0669	A	B	0.15→0.73	both
s_171714_3810	54.7757	-28.6645	A	B	0.02→0.72	DR11
s_90958_9241	59.7052	-46.5695	A	B	0.03→0.70	both
s_176778_4403	50.8798	-27.6486	A	B	0.03→0.70	both
s_170452_14504	54.9726	-29.0757	A	B	0.18→0.65	both
s_84129_16461	62.7892	-48.2719	A	B	0.04→0.62	both
s_88015_3655	65.1247	-47.3087	A	B	0.05→0.62	both
s_88988_8317	62.3367	-47.0669	A	B	0.03→0.62	both
s_167928_8112	52.3902	-29.5927	B	A	0.02→0.95	both
s_84129_17659	62.8043	-48.2286	B	A	0.03→0.93	DR10
s_85093_15882	63.2503	-48.0737	B	A	0.04→0.93	both

Table 6: NEW v3 grade-A candidates from C-vet and their qualification panel verdict (D2; advocate/skeptic/morphology/contaminant, 2-of-N A-rule). DESI-resolution only (no Euclid coverage at these positions): DESI-qualified candidates, not confirmations.

DESI id	C-vet p_{lens}	ranking	panel grade
s_310364_6649	0.87	head	A
s_441355_1111	0.74	head	A
s_124958_10481	0.82	v3blend8	B
s_102952_9916	0.76	v3blend8	B

DESI id	RA	Dec	Euclid	LensJudge	p_{lens}	release
s_77522_7961	63.7748	-50.0119	B	A	0.02→0.91	both
s_180619_719	51.2423	-27.0221	B	A	0.02→0.83	both
s_179345_702	53.6441	-27.2233	B	B	0.03→0.74	both
s_86061_15501	63.6862	-47.8550	B	B	0.03→0.72	both
s_166673_9358	51.7071	-29.7125	B	B	0.03→0.72	both
s_77510_6177	59.0844	-49.9562	B	B	0.02→0.65	DR10
s_80314_12621	59.0815	-49.3650	B	B	0.04→0.65	both
s_78453_11482	63.9706	-49.7278	B	B	0.02→0.65	both
s_84123_2953	60.2940	-48.3238	B	B	0.03→0.65	both
s_88997_602	65.3650	-47.0251	B	B	0.04→0.65	both
s_81263_7679	60.4929	-48.9067	B	B	0.03→0.62	DR10
s_87026_5695	60.5672	-47.6155	B	B	0.02→0.62	DR10
s_88007_7987	62.2189	-47.2949	B	B	0.03→0.62	DR10
s_84125_13400	61.3553	-48.2723	B	B	0.04→0.60	DR11
s_88996_2557	65.0706	-47.0601	B	B	0.02→0.60	DR11
s_167935_8040	54.3727	-29.3848	B	B	0.04→0.60	DR11
s_75664_6939	58.9597	-50.5340	B	B	0.04→0.55	DR11
s_81254_8721	57.2304	-49.0460	B	B	0.03→0.55	DR11
s_88990_5749	63.0160	-47.0193	B	B	0.03→0.55	both
s_176778_10456	51.0181	-27.7804	B	B	0.03→0.55	both
s_76594_8452	62.6149	-50.3258	B	B	0.04→0.50	DR11
s_176791_230	54.4517	-27.8613	C	A	0.03→0.93	both

B Panel-qualified NEW grade-A candidates

Table 6 lists the NEW grade-A candidates surfaced by v3 in the DR10 vetting and their verdict under the campaign qualification panel (advocate/skeptic/morphology/contaminant, with a skeptic veto and a 2-of-N A-rule — the same mechanism that took the v2 campaign's first-pass A/B from 8 to 5). All four survive at ≥B with no contaminant flag, which is cleaner than the v2 campaign. These positions have *no* Euclid coverage, however, so they are DESI-resolution- qualified candidates, not confirmations.

C Reproducibility

All results are reproduced from tracked code and artifacts on branch `reproductions/claudenet-v3`. The script numbering is: `00--91` v1 (members, calibration, combiners, the seven-phase program, tables/figures); `100--165` v2 (NegEval-1M, mining, refit, the DR9 sweep, group-conformal selection);

`200--260` the DR9 qualification campaign; and `300--367` v3 — `300--326` workstream A (mimic metric, bank, retraining, ensemble refit, lens-vs-mimic head), `360--367` deployment (DR10/DR11 parents, scoring, selection, the Euclid cross-validation and the model-progression comparison). The vetting harness lives under `reproductions/lensjudge`. Shared libraries (`_ensemble`, `_train`, `_clib`, ...) reuse the baseline matched-FPR arithmetic verbatim, so every v1/v2/v3 number is directly comparable. The v3 figures and tables in this report are regenerated by `92_make_v3_figures.py` and `92_make_v3_tables.py` from the artifacts under `data/v3/` and `v3/`.

Survey-scale compute ran on NERSC Perlmutter (GPU account `cosmo_g`, CPU `cosmo`); the v3 program used $\mathcal{O}(10^2)$ GPU-hours. Agentic vetting (LensJudge) spent ∼\$53 of a \$100 cap, targeted on the most-promising candidates with an evidence gate. We carry the lineage's honesty discipline throughout: "candidate" never "discovery"; recall of the published catalogues is reported before any new-candidate count; both independent graders are Claude models (different harness/prompt, not statistically independent); the mimic-FPR seed metric is selection-biased and its cross-model reading is corrected against the independent Euclid measure (§1); and DR11 results are reported embargo-aware (internal until the DR11 public release).

References

Euclid Collaboration. Euclid quick data release (Q1): The strong lensing discovery engine. *A&A*, 2025. Q1 Deep Fields; $0.1''$/px VIS.

MNRAS. Contaminant-aware hard-negative augmentation for strong-lens finding. *MNRAS*, 2025. doi:10.1093/mnras/staf327; 11× false-positive reduction at 2.3% true-positive cost.